

Python

1. What is Python?

Python is a language to write programs and automate tasks (very useful in cybersecurity).

2. Variables

Store data.

```
name = "Umar"  
age = 20  
print(name, age)
```

3. Data Types

```
a = 10    # int  
b = 3.5   # float  
c = "hello" # string  
d = True  # boolean
```

4. User Input

```
name = input("Enter name: ")  
print("Hello", name)
```

5. If Condition

```
port = 22  
if port == 22:  
    print("SSH open")
```

6. If-Else

```
port = 80
if port == 443:
    print("HTTPS")
else:
    print("Not HTTPS")
```

7. If-Elif-Else

```
port = 21
if port == 22:
    print("SSH")
elif port == 21:
    print("FTP")
else:
    print("Other")
```

8. Loops

For Loop

```
for i in range(1, 4):
    print(i)
```

While Loop

```
i = 1
while i <= 3:
    print(i)
    i += 1
```

9. Arrays (Lists in Python)

Lists = store multiple values.

```
ports = [21, 22, 80, 443]
```

```
print(ports[0]) # first element
print(len(ports)) # length
```

Loop through list:

```
for p in ports:  
    print("Port:", p)
```

10. Dictionaries (Key-Value)

Very useful in cybersecurity.

```
service = {  
    "port": 22,  
    "name": "SSH"  
}
```

```
print(service["name"])
```

11. Mathematical Operations

```
a = 10
```

```
b = 5
```

```
print(a + b) # addition  
print(a - b) # subtraction  
print(a * b) # multiplication  
print(a / b) # division  
print(a % b) # modulus
```

12. Functions

```
def greet(name):  
    print("Hello", name)
```

```
greet("Umar")
```

13. Try & Except (Error Handling)

```
try:
    num = int(input("Enter number: "))
    print(num)
except:
    print("Invalid input")
```

👉 Prevents program crash

14. Classes (OOP Basics)

```
class Person:
    def __init__(self, name):
        self.name = name

    def show(self):
        print("Name:", self.name)
```

```
p1 = Person("Umar")
p1.show()
```

15. File Handling

```
file = open("test.txt", "r")
print(file.read())
file.close()
```

16. Running System Commands

```
import os
os.system("nmap 127.0.0.1")
```

17. Simple Cybersecurity Example (Port Check)

```
import socket
```

```
target = "127.0.0.1"
```

```
for port in [21, 22, 80]:
    s = socket.socket()
    result = s.connect_ex((target, port))

    if result == 0:
        print("Port open:", port)
    else:
        print("Port closed:", port)

s.close()
```

18. Combining Concepts

```
ports = [21, 22, "abc"]

for p in ports:
    try:
        p = int(p)
        print("Checking port", p)
    except:
        print("Invalid port:", p)
```